

LEE'S SUMMIT, MO, USA

WMO: 720306

Lat: 38.960N

Lon: 94.371W

Elev: 304

StdP: 97.73

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53879

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.3	-12.6	-20.8	0.6	-13.4	-18.8	0.7	-11.7	12.1	2.1	11.1	1.3	3.8	320	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	35.3	24.1	33.6	24.0	32.2	23.7	25.9	32.2	25.2	31.3	24.6	30.3	4.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.7	30.0	23.4	18.9	29.2	22.7	18.1	28.4	82.0	32.1	78.8	31.3	75.8	30.3	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.2	8.3	-18.6	37.5	2.6	2.0	-20.4	39.0	-21.9	40.1	-23.4	41.2	-25.3	42.7		
(5)			-19.1	27.2	2.5	0.7	-20.9	27.6	-22.4	28.0	-23.7	28.4	-25.6	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.4	-0.4	1.0	8.0	13.2	18.8	24.2	26.1	25.0	21.1	14.1	7.6	1.5		
	DBStd	10.71	6.78	6.82	6.47	5.33	4.63	3.17	3.23	3.29	4.10	5.31	5.70	5.97		
	HDD10.0	1132	327	262	115	29	2	0	0	0	21	108	269			
	HDD18.3	2627	582	486	323	164	54	2	0	1	152	322	523			
	CDD10.0	2376	3	9	53	126	274	424	498	464	334	147	36	4		
	CDD18.3	830	0	0	3	12	68	176	240	207	103	21	1	0		
	CDH23.3	7967	0	1	19	93	510	1704	2622	2018	843	152	4	0		
	CDH26.7	3168	0	0	1	14	123	656	1191	862	284	37	0	0		
(14) Wind	WSAvg	4.0	4.5	4.5	4.8	4.7	3.8	3.8	3.3	3.1	3.4	3.9	4.4	4.3		
(15) Precipitation	PrecAvg	1078	35	38	69	105	136	144	103	108	129	96	66	42		
	PrecMax	1525	91	105	238	260	339	276	335	425	409	235	220	124		
	PrecMin	660	0	0	11	9	35	25	4	1	15	4	0	0		
	PrecStd	220	25	26	43	60	70	69	63	79	92	57	48	30		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.6	21.7	25.6	28.8	31.3	35.3	38.2	37.1	33.8	30.0	23.2	18.7		
		MCWB	12.1	12.8	17.1	19.1	21.7	24.4	23.3	23.6	22.9	22.1	15.8	14.6		
	2%	DB	15.2	18.0	23.0	25.9	29.5	33.2	35.8	34.7	31.6	27.3	21.0	15.4		
		MCWB	9.6	11.7	15.6	17.3	21.0	23.8	24.0	23.9	23.1	19.4	14.6	11.2		
	5%	DB	12.2	14.4	20.8	23.9	28.0	32.0	34.0	32.9	29.9	24.6	18.5	12.5		
		MCWB	7.3	9.5	14.3	16.3	20.4	23.7	24.3	23.8	22.5	18.2	12.9	8.8		
	10%	DB	8.9	11.2	18.1	22.0	26.4	30.5	32.3	31.2	28.0	22.5	16.1	10.1		
		MCWB	5.1	6.9	12.6	15.4	19.7	23.0	24.0	23.5	21.3	17.0	11.4	6.7		
	0.4%	WB	14.2	15.2	18.5	20.2	23.7	26.2	27.0	26.7	25.1	22.9	18.1	15.7		
		MCDB	16.6	18.7	24.0	27.4	29.7	31.9	33.4	33.2	30.7	27.9	21.4	18.0		
	2%	WB	10.4	12.6	17.0	18.8	22.5	25.2	26.2	25.7	24.0	21.5	15.6	12.2		
		MCDB	13.8	16.6	21.4	24.1	27.8	31.6	32.5	32.1	29.6	25.4	19.1	14.3		
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	7.8	9.9	15.4	17.5	21.4	24.4	25.4	24.9	23.2	19.9	14.0	9.4		
		MCDB	11.2	14.2	19.4	22.2	26.3	30.2	31.4	31.0	28.3	23.0	17.5	11.9		
	10%	WB	5.1	6.9	13.2	16.1	20.5	23.6	24.7	24.1	22.3	17.9	12.0	6.8		
		MCDB	9.2	11.2	16.9	21.0	25.1	29.1	30.4	29.5	27.1	21.5	15.6	9.8		
	5% DB	MDBR	9.5	9.8	10.3	10.5	9.7	9.7	9.6	9.7	10.2	10.2	9.7	8.9		
		MCDBR	13.5	14.5	12.9	12.6	11.3	10.8	11.1	11.6	10.7	11.9	12.6	12.0		
(36) Mean Daily Temperature Range	5% WB	MCWBR	9.1	9.4	7.4	7.3	5.5	4.6	3.8	4.5	4.6	6.1	8.0	8.8		
		MCDBR	12.5	13.5	10.7	10.9	10.1	10.0	9.7	10.2	9.8	9.9	11.0	11.3		
	5% WB	MCWBR	9.3	9.9	7.3	7.4	5.4	5.0	4.5	4.7	4.7	6.3	8.4	9.7		
		MCDBR	12.5	13.5	10.7	10.9	10.1	10.0	9.7	10.2	9.8	9.9	11.0	11.3		
(40) Clear-Sky Solar Irradiance	taub		0.293	0.305	0.334	0.384	0.425	0.426	0.437	0.437	0.385	0.343	0.302	0.289		
	taud		2.472	2.444	2.417	2.295	2.238	2.269	2.258	2.266	2.362	2.470	2.530	2.509		
	Ebn at Noon		888	923	925	892	858	855	841	832	858	866	872	868		
	Edh at Noon		79	93	106	128	138	134	135	130	111	90	74	71		
(44) All-Sky Solar Radiation	RadAvg		2.20	2.96	3.86	4.92	5.68	6.44	6.42	5.68	4.79	3.40	2.39	1.90		
	RadStd		0.23	0.33	0.34	0.31	0.50	0.50	0.40	0.30	0.32	0.38	0.24	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page